

YAHYE GEDI

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EDUCATION

Carleton University - Bachelor of Computer Science Honours | **CGPA: 11.8/12**

Expected Graduation: April 2027

Artificial Intelligence and Machine Learning Specialization, Minor in Mathematics

- **Relevant Courses:** Data Structures & Algorithms, Object-Oriented Programming, Systems Programming, Web Development, Databases, Linear Algebra, Calculus, Machine Learning, Artificial Intelligence, Discrete Mathematics, Software Engineering, Data Science & Statistics
- **Awards:** Entrance Scholarship, Dean's Honour List, Department Awards, Golden Key Inductee, Community Involvement

WORK EXPERIENCE

Carleton University

Sep 2025 - Present

Undergraduate Teaching Assistant - COMP 3106: Introduction to Artificial Intelligence | Ottawa, ON

- Incoming Teaching Assistant for an Intro to AI class at Carleton University this upcoming Fall semester.

Ajile Light Industries

Jan 2025 - Sep 2025

Software Developer Intern | Ottawa, ON

- Led calibration feature development for the MultiDepthScan system, enabling crosshair reprojection and sphere-based alignment across 8 depth scans for high-precision calibration.
- Designed and implemented a mock turntable simulation system to emulate 360° point cloud capture, accelerating testing workflows without hardware dependency.
- Developed features to improve traceability in the Back Office, including product disassembly tracking and return-item reporting, ensuring consistency and accountability across workflows.
- Built GUI test cases for the DMD Controller Interface, validating key functions (project loading, triggering, sequence creation) and integrating with GitLab CI/CD to improve software reliability.

Carleton University

May 2024 - Sep 2024

Undergraduate Teaching Assistant - COMP 1406: Introduction to Computer Science II | Ottawa, ON

- Instructed 120+ students in key programming topics including OOP, loops, recursion, and file I/O through weekly tutorials, live demonstrations, and guided exercises.
- Delivered one-on-one and group support for both in-person and remote learners via Zoom, tailoring explanations to diverse learning styles and improving course engagement.

KEY PROJECTS

Neural Network Image Classifier

Python, TensorFlow, Keras, Pandas, Numpy

- Designed and trained a **hybrid CNN-GRU model** for sequential image classification, combining spatial feature extraction with temporal learning for improved accuracy.
- Achieved **95%+ validation accuracy** over 40 epochs using the Adam optimizer with adaptive learning rates and effective data splitting strategies.
- Evaluated model performance on test data, achieving **90% accuracy** and demonstrating strong generalization across diverse handwritten digit sequences.

Carleton University AI Email Classifier

Python, APIs, Pandas, Git, Github

- Built a CMAIL-based email classification system using TF-IDF vectorization and K-means clustering to categorize incoming emails into user-friendly groups.
- Validated clustering performance with **high silhouette scores**, producing clean mappings for categories like "Events" and "Feedback."
- Developed a hybrid prediction engine combining keyword-based rules with distance-based classifiers, improving classification accuracy across 2500+ real emails.

Relational Database Optimization Engine

Python, SQL, Postgres, pgAdmin, UML, psycopg, Git, Github

- Built a high-throughput data ingestion pipeline using **psycopg**, reducing PostgreSQL import time for large datasets by **80%**.
- Implemented robust error handling for **500K+ records**, ensuring reliability during bulk loads with minimal manual intervention.
- Optimized queries by applying indexing and table partitioning strategies, reducing execution times by **up to 95%**.

Meridian Health Tracker

C++, QtCreator, VirtualBox, SSH, Git, Github

- Engineered a C++ user management system with dynamic data structures for user registration, session tracking, and historical data logging.
- Simulated real-time health data acquisition with features like sensor readiness validation, battery monitoring, and asynchronous device scanning.
- Implemented a scalable historical data archive, combining efficient memory usage with secure and structured user-specific data storage.

SKILLS

Languages: Python, C, C++, C#, Java, JavaScript, TypeScript, HTML, CSS, SQL, Go

AI/ML Frameworks: TensorFlow, PyTorch, Keras, Scikit-learn, Keras, Pandas, Numpy, AWS, Matplotlib

Networking & Embedded Systems: TCP/IP, WebSockets, FPGA, SPI, I2C, GNU toolchain (GCC, GDB, Makefiles)

Databases: PostgreSQL, MySQL, SQLite, MongoDB, Firebase, Redis, DynamoDB, PgAdmin,

Web Development: React, Node.js, Next.js, Express.js, Django, Tailwind, XML, Flask, REST APIs

Tools & Platforms: Git/GitHub, Linux, Jira, Agile, VsCode, VirtualBox, Raspberry Pi, APIs, Bitbucket, Bash